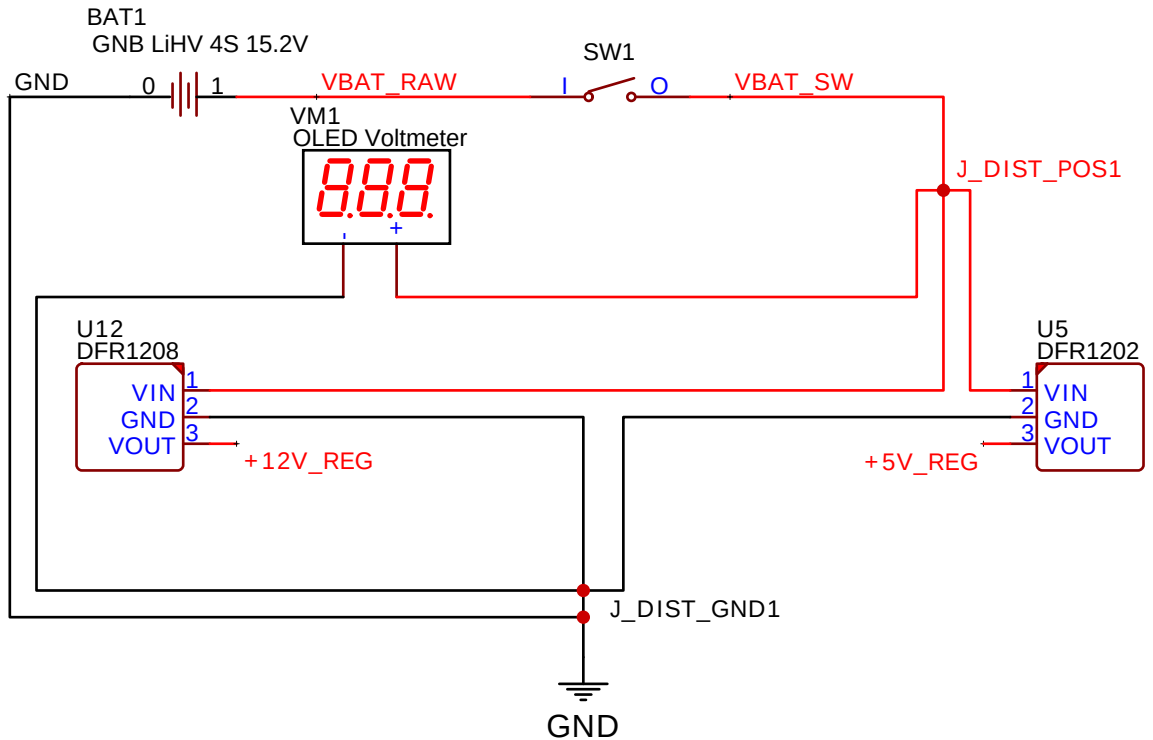
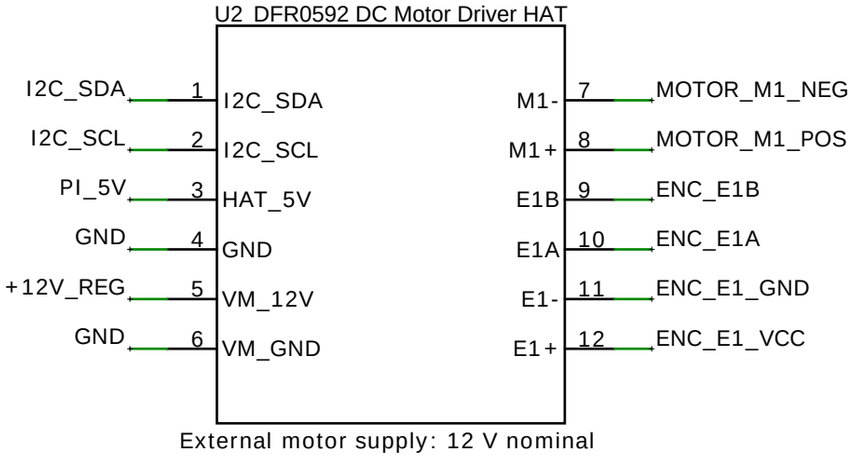
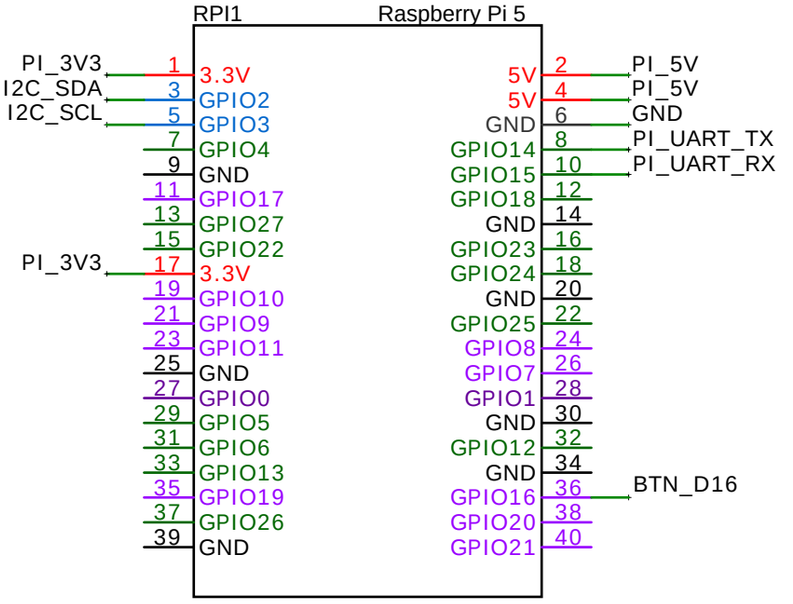
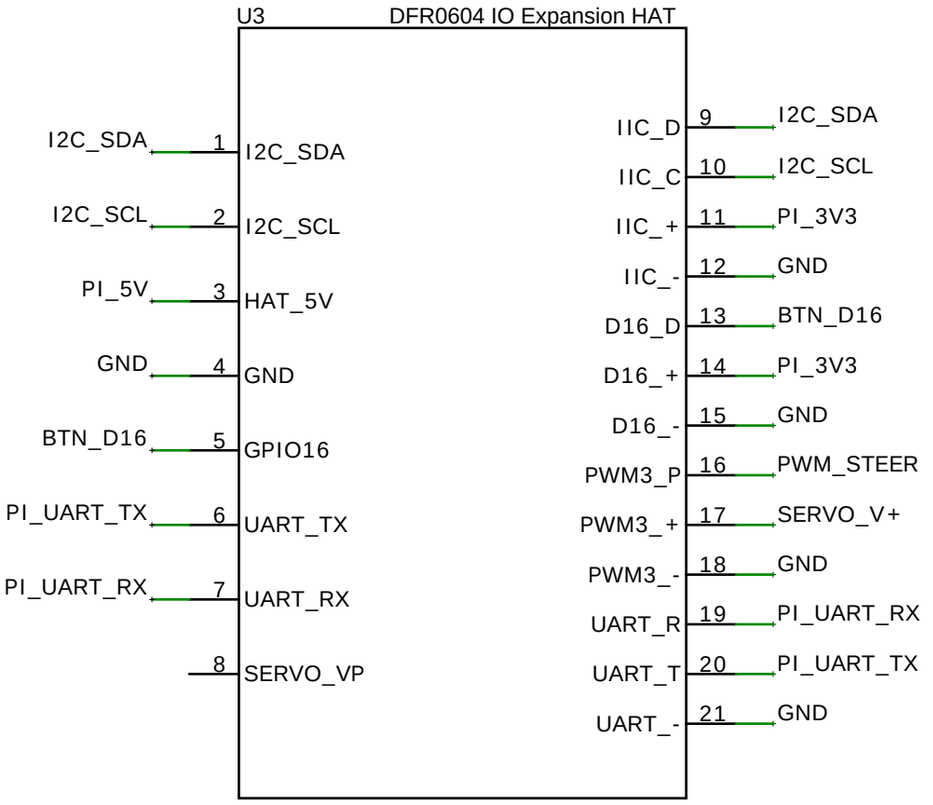


TITLE: POWER DISTRIBUTION		REV: D1
EasyEDA	Company: Team YBR-SUNFLOWER	Sheet: 1/3
	Date: 2026-10-23 Drawn By: peachvscode	

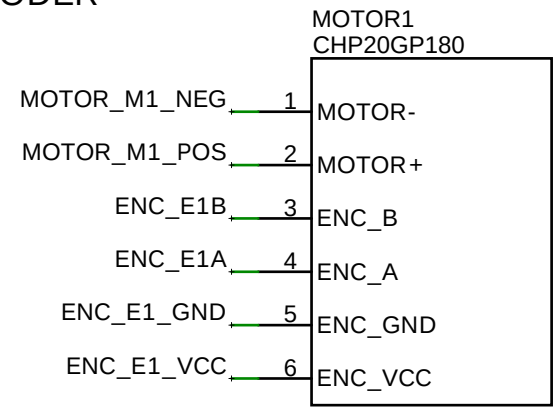






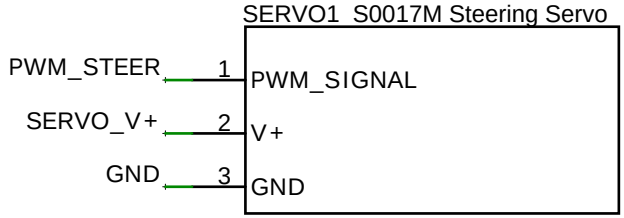
TITLE: MAIN CONTROLLER & HAT INTERFACES		REV: D1
EasyEDA	Company: Team YBR-SUNFLOWER	Sheet: 2/3
	Date: 2026-10-23 Drawn By: peachvscode	

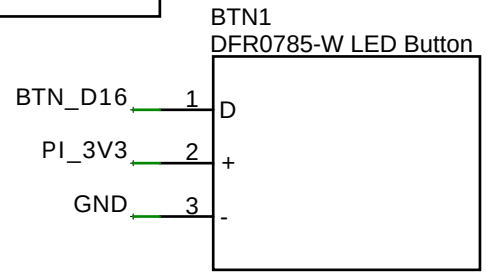
DC MOTOR & ENCODER



CHP-20GP-180 | 12 V geared DC motor
Motor terminals: M1+ / M1-
Encoder: E1A / E1B quadrature channels
Encoder supply: DFR0592 E1+ / E1-
DFR0592 logic supply: 5 V nominal; encoder powered from E1+/E1−

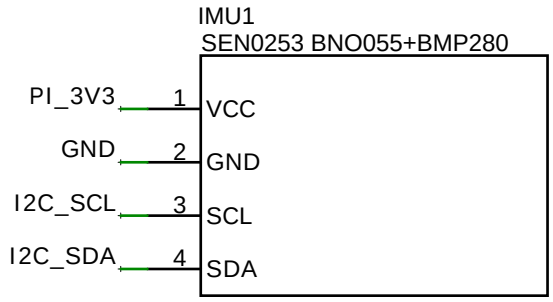
STEERING SERVO & USER INTERFACE

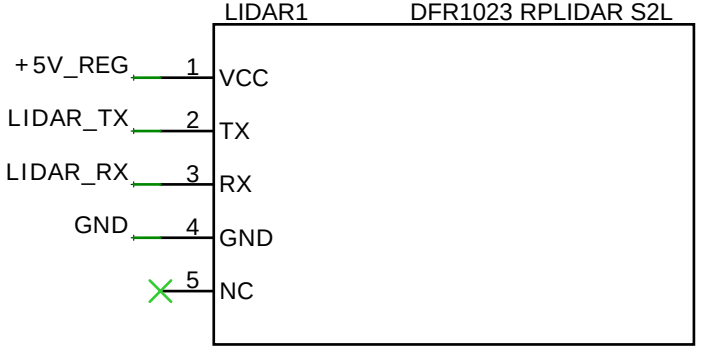




S0017M steering servo, PWM channel 3
Rated supply: 4.8-6.0 V
PWM V+: 5 V nominal via DFR0604 (VP unpowered)

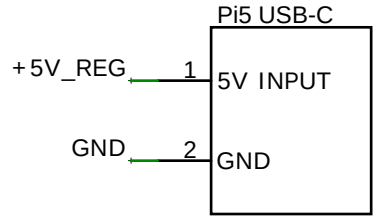
PERCEPTION SENSORS





CAM1 OV5647 Fisheye Camera
Interface: MIPI CSI ribbon
Host: Raspberry Pi 5
Port: CAM1

RASPBERRY PI POWER & ACCESSORIES



DFR1202 -> USB-C cable -> Raspberry Pi 5

FAN1 Raspberry Pi 5 Active Cooler
Connection: dedicated Pi 5 FAN header
Power/control: via official FAN connector

TITLE: SENSORS, MOTOR, STEERING		REV: D1
EasyEDA	Company: Team YBR-SUNFLOWER	Sheet: 3/3
	Date: 2026-10-28 Drawn By: peachvscode	